

Coding Agent Memory Post-training

Unlocking the Memory Potential of Pre-trained File Operations
for Long-Horizon Tasks via Reinforcement Learning

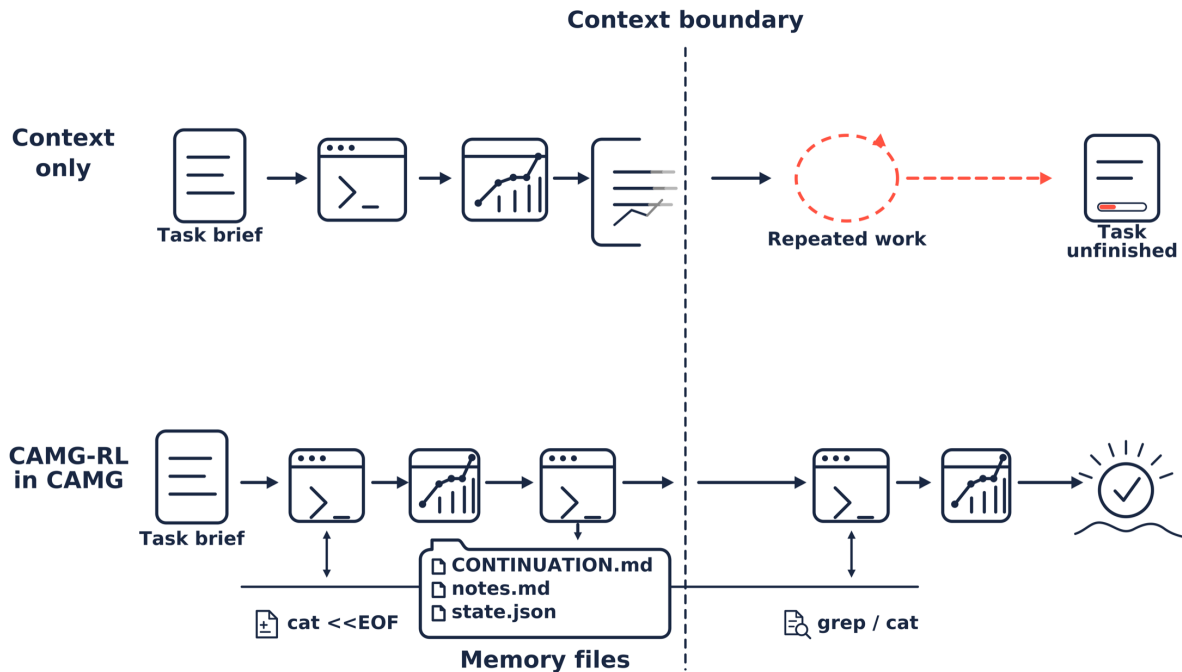
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JD.com

Paper · [arXiv:2609.34422](https://arxiv.org/abs/2609.34422)

Project · liruiluo.github.io/agentmemorygym

Long tasks need state that outlives context



Useful state survives a context replacement.

MOTIVATION A bounded active context is not a durable workspace.

Can RL unlock the memory prior in file operations?

PRE-TRAINING PRIOR

**Shell commands,
files, search,
and revision are
familiar actions.**

OPEN QUESTION

**Can downstream task
reward
turn that prior into
long-horizon memory?**

RESEARCH QUESTION Learn memory behavior through the task objective, without inventing a new memory tool vocabulary.

CAMG spans four native task worlds

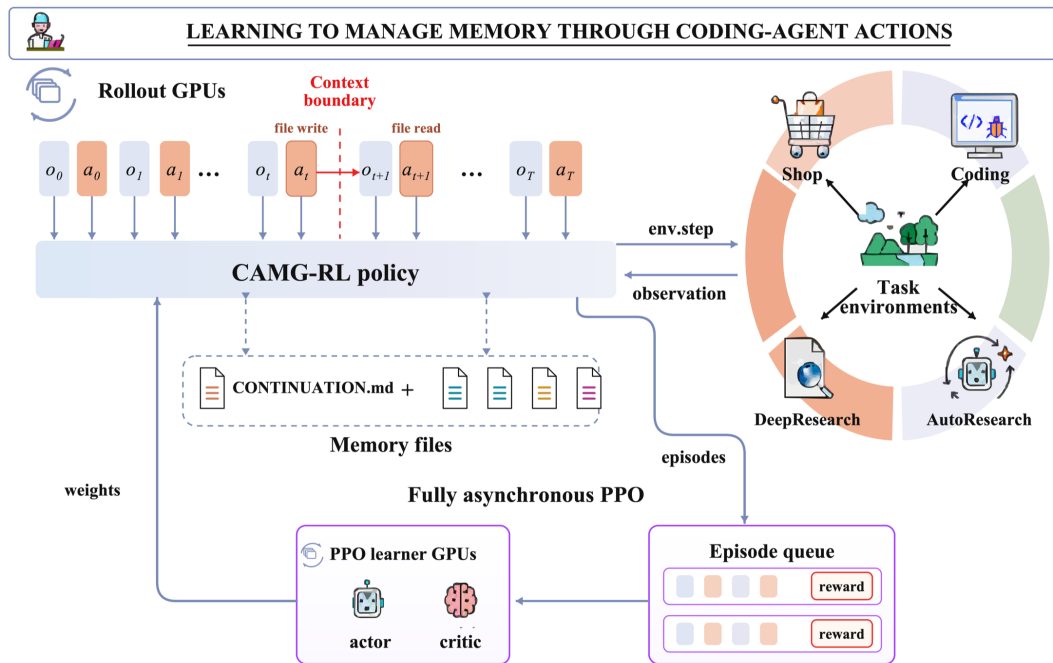
The image displays four distinct task environments within the CAMG framework, each with a blue header and a white body.

- Shop:** Features a sequence of tasks. The 'Earlier' task involves purchasing 1 of 6 phone cases, with options for 'Black silicone case' (approved listing 1) and 'Space Gray silicone case' (approved listing 2). A green checkmark indicates 'Customer confirms listing 1.'. The 'Later' task involves purchasing 4 of 6 window curtains, with a prompt to 'Use what you learned from my earlier choices for the color.' and options for 'Black blackout curtain' and 'Light Gray sheer curtain'.
- Coding:** Presents a problem: 'Issue: SectionWrapper.items() reverses items and swaps keys with values'. It includes a 'Reproduction' code block with a Python snippet that iterates over a dictionary and prints its items. Below the code, it states '5 issue tests · 23 regression tests' and provides a button to 'Submit a repository patch'.
- DeepResearch:** Starts with a 'Question' about the 2024 Puerto Rican general election. It includes a 'Search' section with a query '2024 Puerto Rico governor vote share independence alliance' and a 'Visit' section with a link '2024 Puerto Rican general election'. A green text prompt at the bottom says 'Answer using retrieved evidence.'.
- AutoResearch:** Focuses on 'Seattle House Price Prediction'. It provides data for 'train.csv' (2,016 rows) and 'test.csv' (505 rows) with columns for id, size, zip, and price. It includes a table of sample data for both sets. Below the tables are buttons for 'Target · price' and 'Metric · RMSE ↓'. It also lists 'Features: beds · baths · size · lot_size · zip_code' and a button to 'Write submission.csv and submit once.'.

- four environments: Shop, Coding, DeepResearch, and AutoResearch
- native task interface + shell + episode-persistent workspace
- one shared policy and downstream reward path

CAMG The Gym holds task semantics fixed while exposing a common file-backed memory substrate.

One policy learns across all four environments



native task action



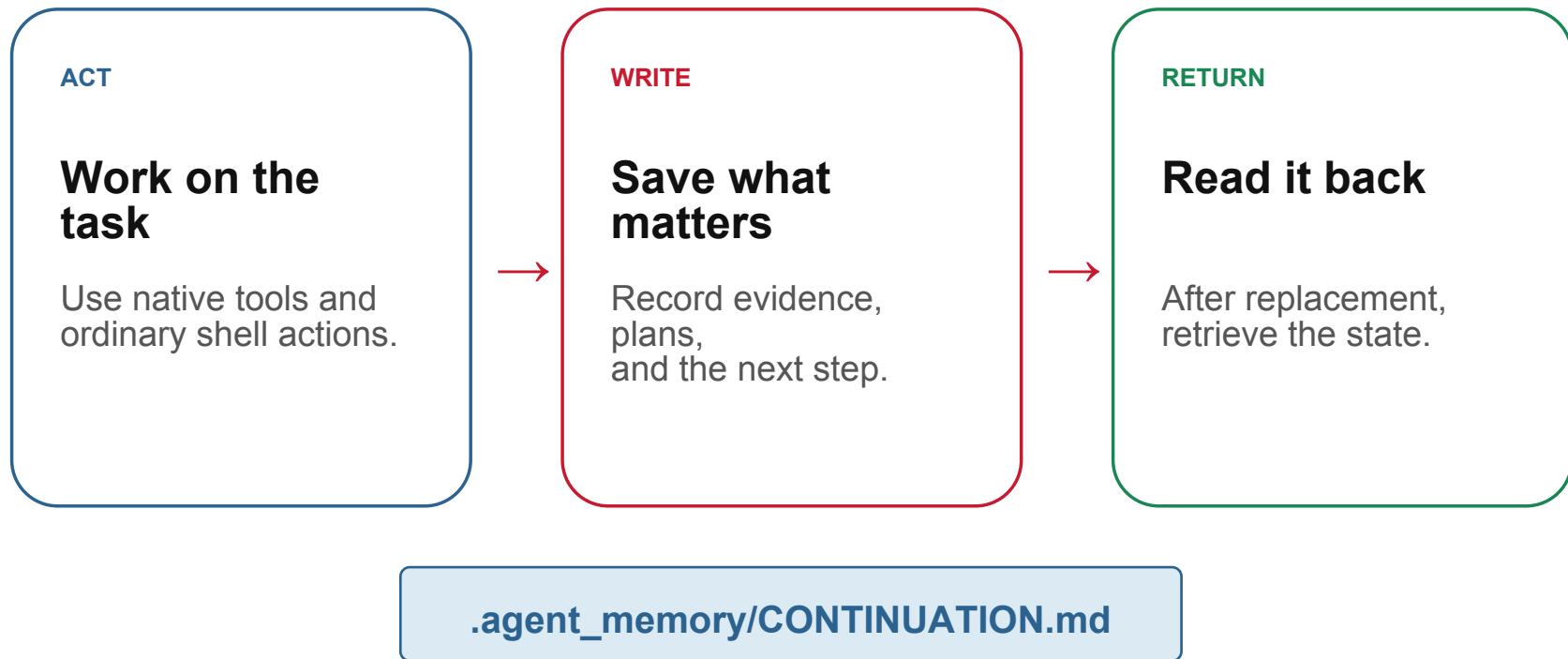
filesystem action



shared task reward

SYSTEM Fully asynchronous training absorbs variable episode durations while keeping a single actor and critic.

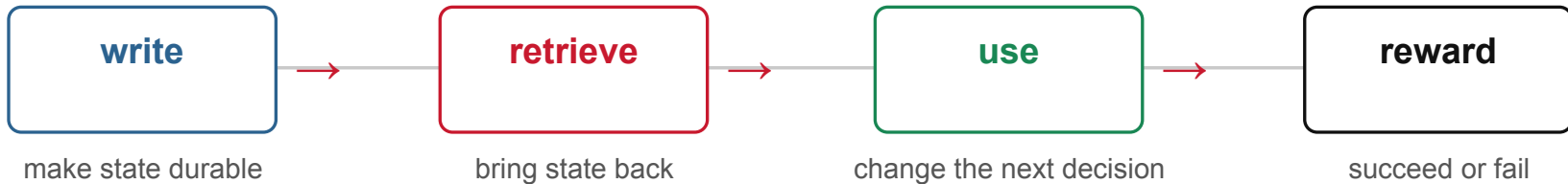
Files carry reasoning across context boundaries



MEMORY AS REASONING No separate hidden memory channel: editable files carry plans, evidence, and intermediate findings.

Task reward trains the whole memory chain

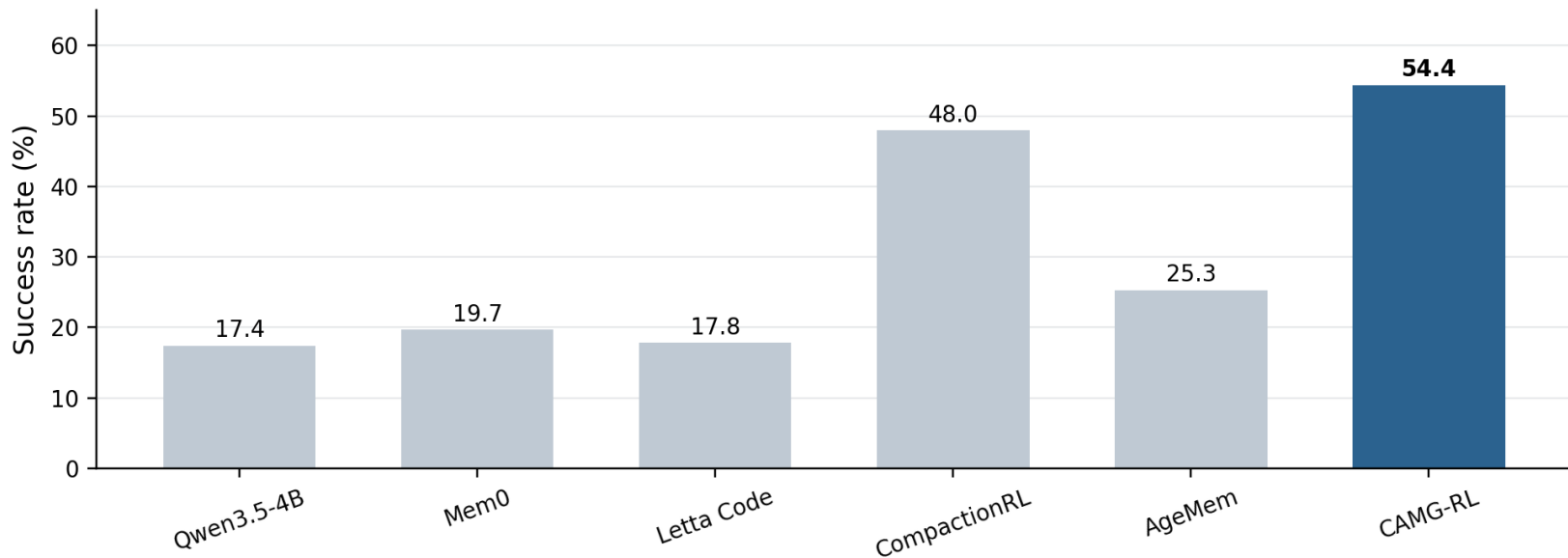
One task reward



One shared credit path

Native task reward provides the learning signal.

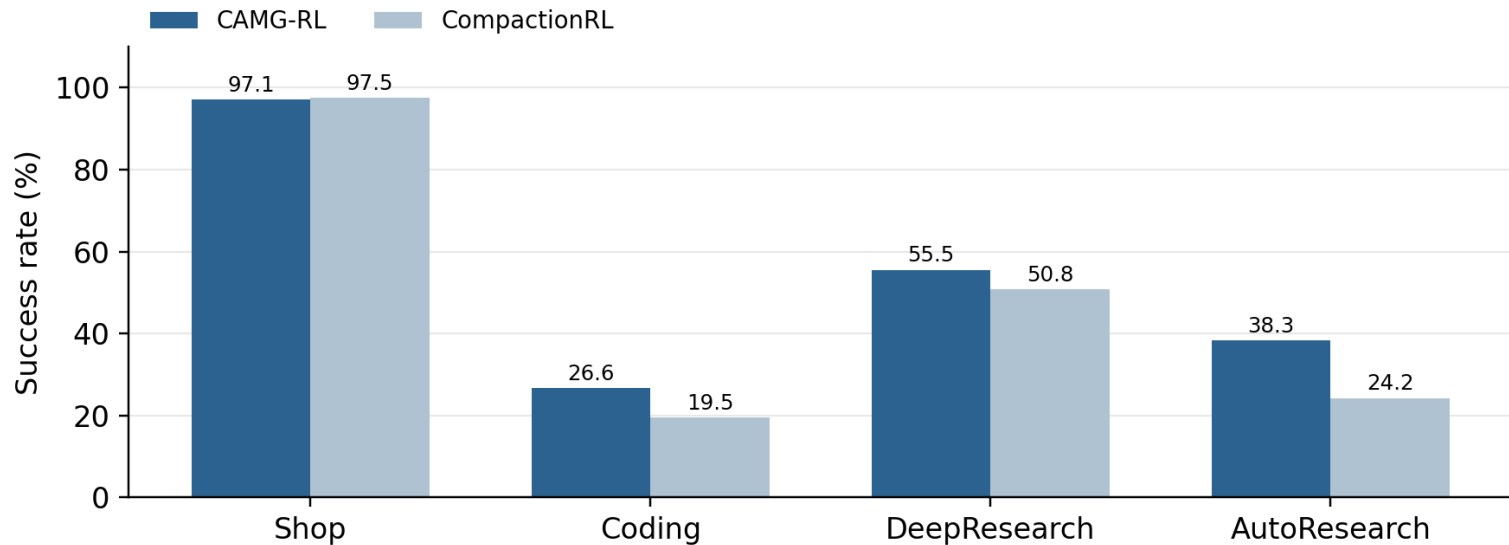
CAMG-RL achieves 54.4% average success



+6.4 percentage points over CompactionRL

TABLE 2 128 test tasks per environment; matched decoding, action budgets, runtimes, and graders.

The largest gains are in Coding and AutoResearch

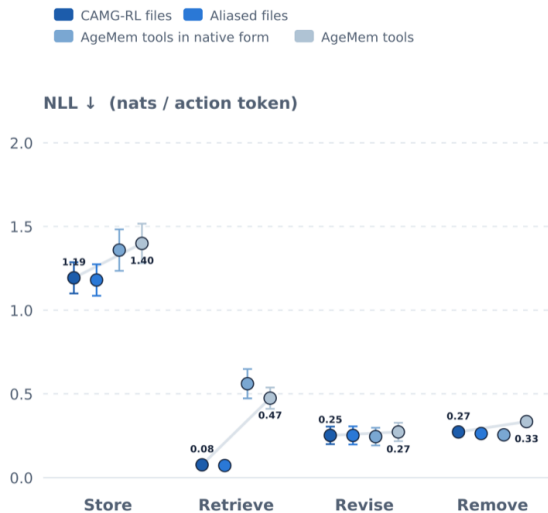


- Shop: both methods nearly saturate the task
- Coding: 26.6% vs. 19.5% for learned compaction
- AutoResearch: 38.3% vs. 24.2% — the largest absolute gap

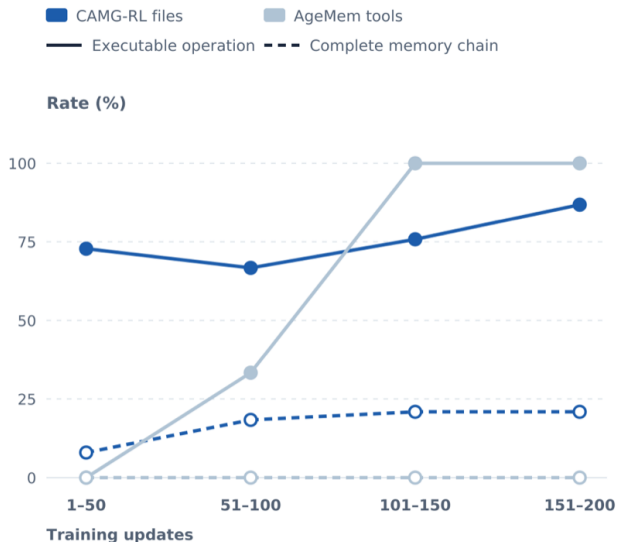
RESULT A single policy learns one memory behavior while each environment keeps its native task semantics.

File operations build on a measurable pre-training prior

(a) Frozen-base action surprisal



(b) Memory behavior acquisition



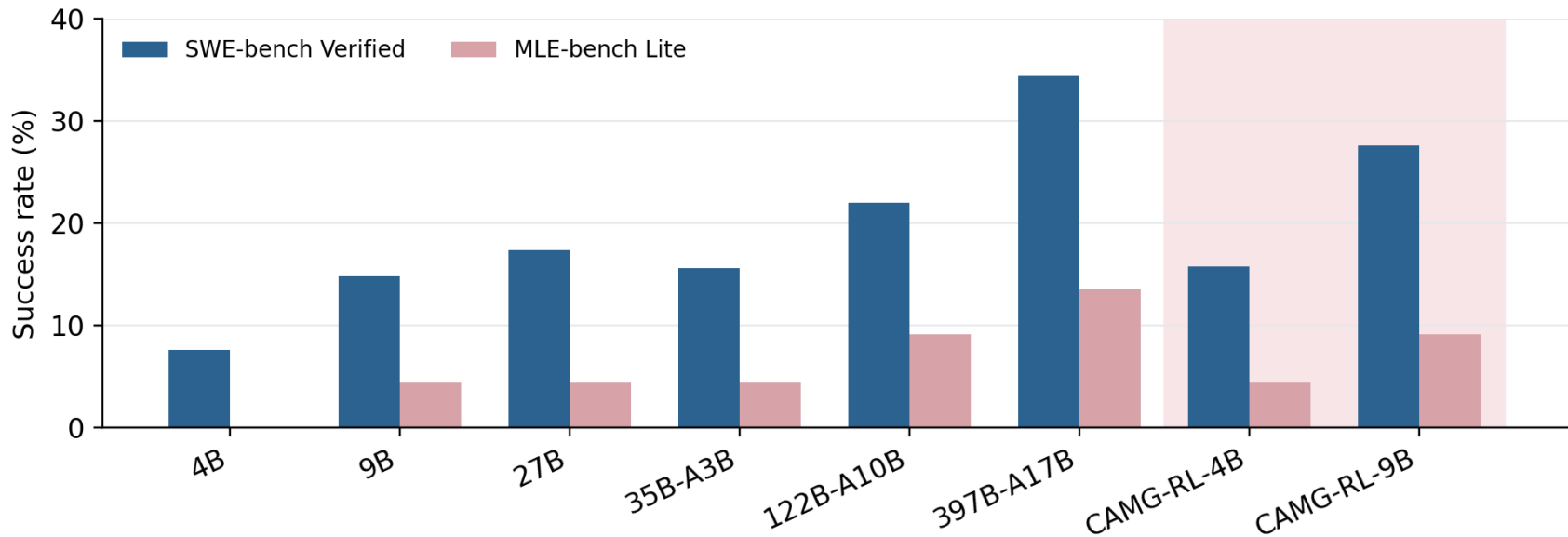
(c) Base-to-trained policy shift



code pre-training supplies the interface prior

INTERFACE PRIOR Training mainly has to discover when to write, revise, retrieve, and reuse — not a new action language.

Learned memory transfers beyond the training Gym



CAMG-RL-4B: 15.8 / 4.5

CAMG-RL-9B: 27.6 / 9.1

EXTERNAL TRANSFER The 4B and 9B policies are competitive with much larger frozen Qwen3.5 models on the paired benchmarks.

Memory-action credit and general files both matter

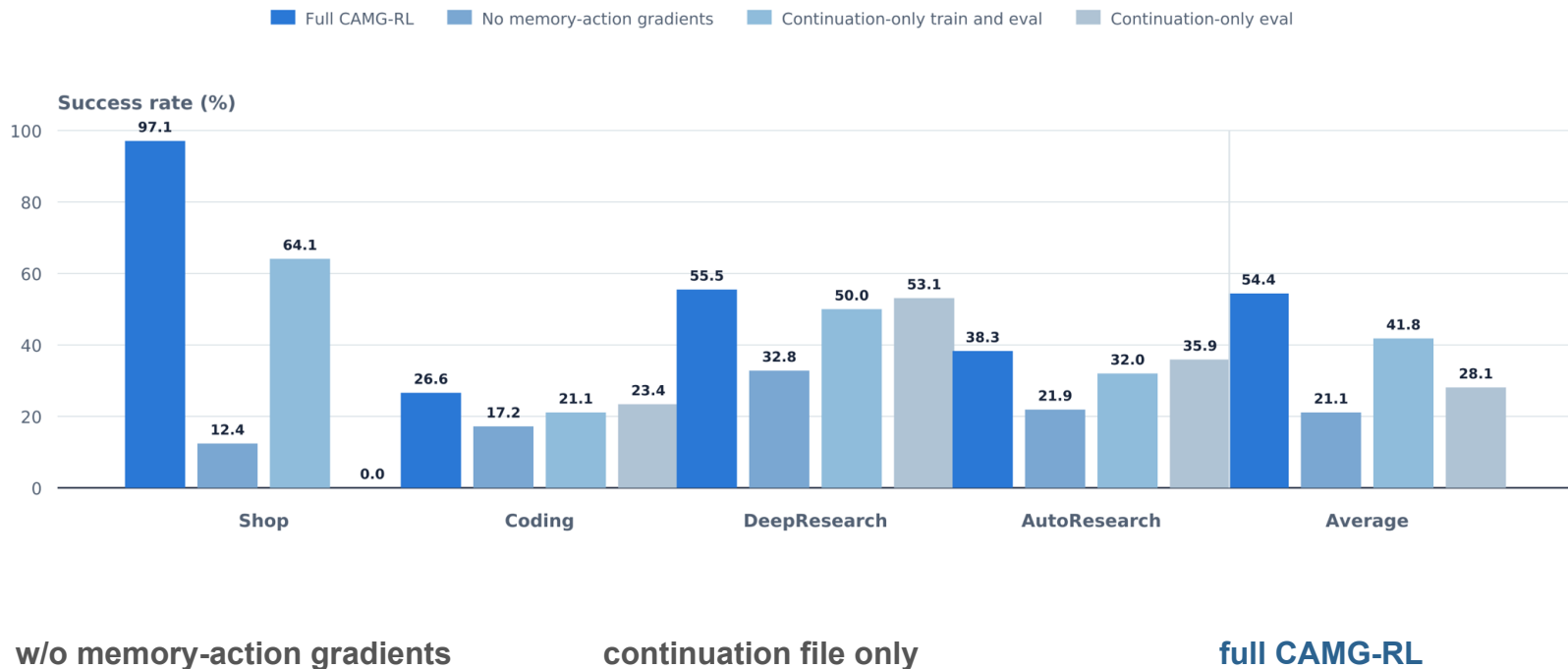


FIGURE 4 Average success: 54.4% full; 21.1% without memory-action gradients; 41.8% when retrained with continuation-only memory.

Changing saved memory content reduces later success

Intervene at the first context replacement

Continue the trajectory after replacing the saved files.

Blank memory

-14.1 pts

Task-mismatched memory

-11.5 pts

Average success-rate change from the matched original-memory condition

A coding episode makes the memory chain visible

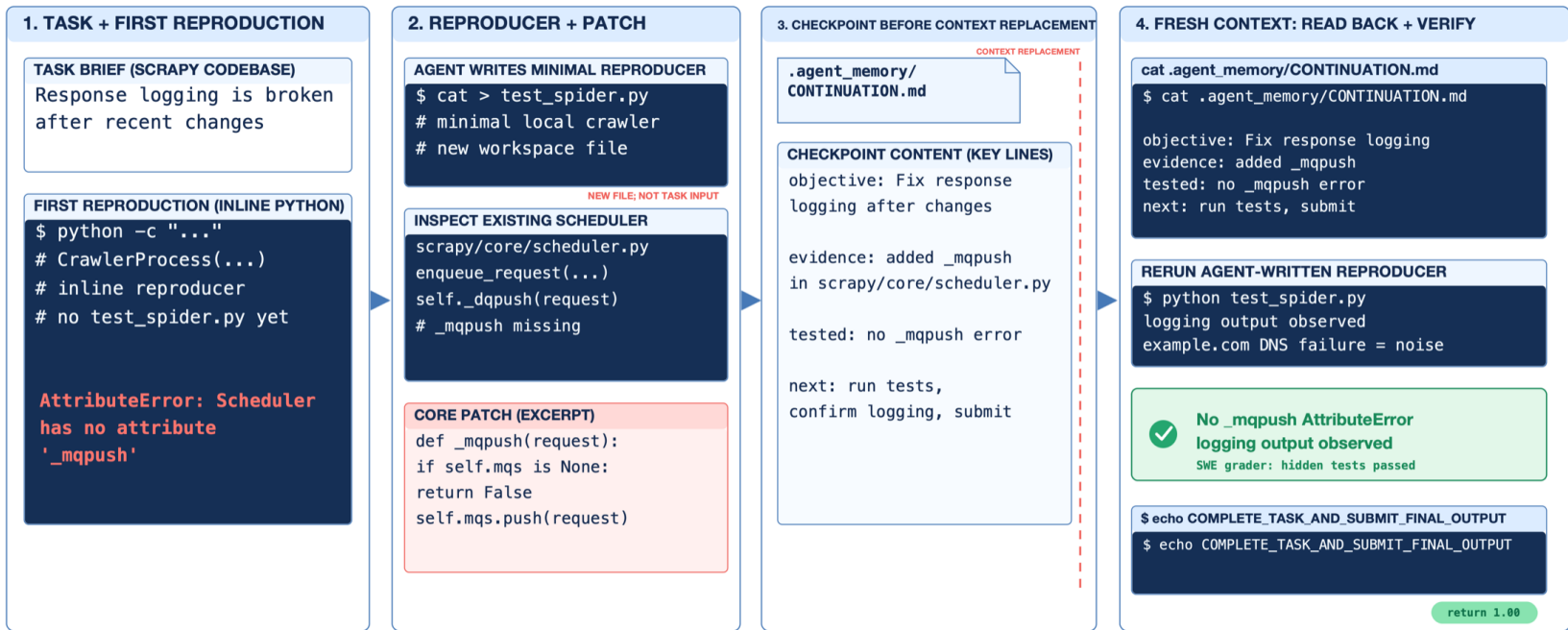


FIGURE 5 · TRAINING CASE Reproduce, patch, save before context replacement, read back, verify, and submit.

Takeaways

- 01** CAMG turns long-horizon memory into a testable, multi-domain RL environment suite.

- 02** CAMG-RL learns to use ordinary files as memory through downstream task reward.

- 03** A small post-trained policy can transfer this behavior to external agent benchmarks.

Files are not just artifacts. They are memory.